

Stanford University's School of Engineering
Computer Science
Computational Biology Track
2024-2025 Program Sheet

Final version of program sheet is due to the department one month prior to the last quarter of senior year.

Follow all requirements as stated for the year of the program sheet used.

Name: Patrick Walsh	SUID #: 6649802	
Phone: 2629515647	Email: walshp26@stanford.edu	
Today's Date: 8/31/2025	Month/YrB.S. expected: June 2026	

Mathematics and Science Requirement

Mathematics and Science Requirement							
Dept	Course	Title	Transfer/AP Approval by SoE			Unit	Grade
			P if Transfer	SoE Initials	Date		
Mathematics (20 units minimum)							
MATH	19	Calculus (see note 1)		AP CreditAU/2310T			
MATH	20						
MATH	21						
MATH	51					5	A-
CS	103	Mathematical Foundations of Computing				5	A
CS	109	Introduction to Probability for Computer Scientists				5	A
Mathematics Unit Total (20 units minimum)						25	

Biology (12 units minimum; take BIO or HUMBIO sequence)

BIO	Any 3 of 82, 83, 85, 86	Genetics/Biochem & Mol Biol/Evolution/Cell Biology					
OR							
HUMBIO	2A, 3A, 4A	Genetics, Evolution & Ecology/Cell & Dev/Human Org				5	A+,A,A
<i>Science Unit Total (12 units minimum)</i>						15	
<i>(32 units min. Math/Sci combined)</i>						40	

Technology in Society Requirement (1 course req'd; see note 7 for WIM options)

HUMBIO	174	A TiS course must be on the SoE-approved list the year you take it				3	A
--------	-----	--	--	--	--	---	---

Engineering Fundamentals (8 units minimum)

CS	106B	Programming Methodology				5	A-
BIOE	80	Elective (see note 2)				4	A+
<i>Engineering Fundamentals Total (8 units minimum)</i>						9	

NOTES

- * All courses listed on this form can be included under only one category. There is no double-counting.
- * All courses listed on this form must be taken for a letter grade (unless taken Spring 2019-20, and Aut-Sum 2020-21)

- * This printed form must be signed by the departmental representative (SSO), with changes petitioned (see UGHB, Petitions page) and initialed/dated by SSO.
- * Minimum Grade Point Average (GPA) for all courses in ENGR Fundamentals and CS Core, Depth, and Senior Project (combined) is 2.0.
- * Students without prior programming experience should first take CS106A. The major otherwise requires at most 95 units.
- * **AP (or IB/GCE) test credit** for use in the major can be confirmed by checking your MAP record in the Bachelor of Science Mathematics, Science, and/or Fundamentals categories.
- * Transfer credits in Math, Science, Fundamentals, & TiS must be approved by the SoE Dean's office. Transfer credit information and SoE & departmental petitions are available at <https://ughb.stanford.edu/transfers-ap-exceptions>. Transfer credits in CS Core, Depth and Senior Project must be approved by the Computer Science office.
- * Courses must be taken for the number of units on the Program Sheet. CS 103, 106B, 107, 109, 111 and 161 must be taken for 5 units.
- (1) MATH 19/20/21 or equivalent (10 units AP BC, or transfer, with placement into MATH 51/CME 100) is acceptable. If 6-8 units AP or IB credit are used, must take Math 21 (21 may not be skipped using Math Diagnostic Placement results).
- (2) One course required; may not be any CS 106. See ENGR Fundamentals Courses list at ughb.stanford.edu

CS Computational Biology Track Program Sheet (continued)

CS Computational Biology Track Core, Depth, and Senior Project (43 units minimum)

Be advised: no course may be listed twice on the sheet; no double-counting.

Dept	Course	Title	Transfer/Deviation Approval by Dept			Unit	Grade
			P if Transfer	Dept Initials	Date		
Core (15 units minimum)							
CS	107 or 107E	Computer Organization and Systems				5	A-
CS	111	Operating Systems Principles				5	A
CS	161	Design and Analysis of Algorithms				5	A
Depth (25 Units minimum)							
CS	173A	Foundations of Computational Human Genomics				AU/25	
CS	221	Artificial Intelligence: Principles and Techniques				4	A
CS	145 or 246	Databases, Mining Massive Datasets				AU/25	
CS	142, 147L, or 448B	Web Applications, Data Visualization				AU/25	
BIOMEDIN	219	Restricted Elective (see note 3) Comp Bio				3	A+
CS	124	Restricted Elective (see note 4) AI				4	A
CS	279	Restricted Elective (see note 5) Quant Tools				3	A
CHEM	141	Restricted Elective (see note 6) Application Areas				4	A
Seior Project (3 Units Minimum)			Total depth units (25 minimum) 33				
CS		At least 3 units of 191, 191W, 194, 194H, 194W, 210B, or 294 (see note 7)					
Computer Science Core and Depth Total (43 units minimum)							

Program Approvals

Departmental (Staff Only)

Printed Name: _____

Date: _____

Signature: _____

School of Engineering (No action required-office use only)

Printed Name: _____

Date: _____

Signature: _____

NOTES (continued from page 1)

- (3) One course selected from CS 279, 371; BIOMEDIN 210, 214, 215, 217, 219, 220, 222, 260, 273B; IMMUNOL 207
- (4) Pick one course from:
Area I) AI Methods: 224R, 224W, 228, 229, 229S, 234, 238; Area II) Natural Language Processing: CS 124, 224N, 224V; Area III) Vision: CS131, 231N
- (5) One course selected from: CS 147, 148, 154, 166, 168, 185, 230, 248A, 353; BIO 183, 187; BIOC 241; BIOMEDIN 248; EE 263, 364A; MS&E 152, 252; STATS 141, 202, 203, 205, 206, 211, 315A, 315B; ChemEng 150; AppPhys 294;
or a class from: CS 142, 147L, 448B, note 3, or a distinct area from note 4
- (6) One course selected from: BioE 220; BIODS 295; ChemEng 150, 174; GENE 211; HUMBIO 151R; ME 281; AppPhys 294; Bio 81, 82, 83, 84, 85, 86, 112, 214, 230; Chem 31A, 31B, 31M, 33, 141, 143, 171; BIOC 241; Dbio 210; Surg 101, or a class from note 3
- (7) The WIM requirement may be met by taking CS 181W or 182W as a Technology in Society course or through the Senior Project course (191W, 194W, or 210B only).